INCH-POUND

MIL-M-28787/433
6 August 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY TIMING MODULE, KEY CODE JEQ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic memory timing module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.434 failures per million operating hours for class I or 3.250 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JEQ.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B5	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Power supply stress	IV	Q	---------- Module shall pass all remaining tests ----------						
D1	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D2	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D3-D6	Ibd	V	S	0 0	0.1	0.0	0.1	0 0	0 1	mA
D7	Ibd	V	S	-0 58	0.78	-0.58	0.78	-0 58	0 78	mA
D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D9	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D10	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D11	Ibd	V	S	0 0	0.3	0.0	0.3	0 0	0 3	mA
D12, D13	Ibd	V	S	-0 58	0.68	-0.58	0.68	-0 58	0 68	mA
D14	Ibd	V	S	-0.58	0.78	-0.58	0.78	-0.58	0.78	mA
D15	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D16, D17	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D18	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D19-D25	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D26	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D27	Ibd	V	S	-0.58	0.68	-0.58	0.68	-0.58	0.68	mA
D28	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D29	Ibd	V	S	-0 58	0.78	-0.58	0.78	-0 58	0 78	mA
D30, D31	Ibd	V	S	0 0	0.1	0.0	0.1	0 0	0 1	mA
D32, D33	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D34	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D35	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D36	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D37	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D38, D39	Ibd	V	S	0.0	0.5	0.0	0.5	0.0	0.5	mA

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 95 ± 5°C at		Units
				Min	Max	Min	Max	Min	Max	
D40-D42	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D43-D45	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D46	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D47	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D48	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D49	Ibd	V	S	0.0	1.1	0.0	1.1	0.0	1.1	mA
D50	Ibd	V	S	0.0	0.0	0.0	0.0	0.0	0.0	mA
D51	Ibd	V	S	0.0	0.5	0.0	0.5	0.0	0.5	mA
D52,D53	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D54	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D55	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D56	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
E1,E2	Ids	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E3,E4	Ids	VI	S	-90.0	-12.0	-90.0	-12.0	-90.0	-12.0	mA
E5-E10	Ids	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E11	Ids	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E12,E13	Ids	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E14	Ids	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E15-E22	Ids	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E23	Ids	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E24	Ids	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E25	Ids	VI	S	-975.0	-300.0	-975.0	-300.0	-975.0	-300.0	mA
E26	Ids	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E27,E28	Ids	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E29	Ids	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E30,E31	Ids	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F2	Iih	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
F3,F4	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F5,F6	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F7	Iih	VII	X	-28.3	-14.3	-28.3	-14.3	-28.3	-14.3	mA
F8	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class J 4/ 0 ± 5°C 60 ± 5°C		Class H 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F9	11h	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F10	11h	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
F11	11h	VII	X	-1.8	0.08	-1.8	0.08	-1.8	0.08	mA
F12,F13	11h	VII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
F14	11h	VII	X	-28.3	-14.3	-28.3	-14.3	-28.3	-14.3	mA
F15	11h	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F16,F17	11h	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F18	11h	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F19-F25	11h	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F26	11h	VII	X	-1.0	0.06	-1.0	0.06	-1.0	0.06	mA
F27	11h	VII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
F28	11h	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F29	11h	VII	X	-28.3	-14.3	-28.3	-14.3	-28.3	-14.3	mA
F30	11h	VII	X	-1.0	0.02	-1.0	0.02	-1.0	0.02	mA
F31	11h	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F32,F33	11h	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F34	11h	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F35	11h	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F36	11h	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F37	11h	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F38	11h	VII	X	-3.0	0.1	-3.0	0.1	-3.0	0.1	mA
F39	11h	VII	X	-7.0	0.04	-7.0	0.04	-7.0	0.04	mA
F40-F42	11h	VII	X	-1.0	0.02	-1.0	0.02	-1.0	0.02	mA
F43,F44	11h	VII	X	-0.6	0.07	-0.65	0.07	-0.65	0.07	mA
F45	11h	VII	X	-0.65	0.02	-0.6	0.02	-0.6	0.02	mA
F46	11h	VII	X	-0.6	0.06	-1.8	0.06	-1.8	0.06	mA
F47	11h	VII	X	-1.8	0.04	-1.2	0.04	-1.2	0.04	mA
F48	11h	VII	X	-1.2	0.09	-3.2	0.09	-3.2	0.09	mA
F49	11h	VII	X	-3.2	0.07	-2.6	0.07	-2.6	0.07	mA
F50	11h	VII	X	-2.6	0.16	-4.8	0.16	-4.8	0.16	mA
F51	11h	VII	X	-4.8	0.1	-3.0	0.1	-3.0	0.1	mA
F52	11h	VII	X	-3.0	0.02	-0.6	0.02	-0.6	0.02	mA
F53,F54	11h	VII	X	-0.6						mA

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F55	11h	VII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
F56	11h	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F57	11h	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G1	111	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G2	111	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
G3,G4	111	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G5,G6	111	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G7	111	VIII	X	-28.3	-14.3	-28.3	-14.3	-28.3	-14.3	mA
G8	111	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G9	111	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G10	111	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
G11	111	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G12,G13	111	VIII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
G14	111	VIII	X	-28.3	-14.3	-28.3	-14.3	-28.3	-14.3	mA
G15	111	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G16,G17	111	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G18	111	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G19-G25	111	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G26	111	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G27	111	VIII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
G28	111	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G29	111	VIII	X	-28.3	-14.3	-28.3	-14.3	-28.3	-14.3	mA
G30	111	VIII	X	-1.0	0.02	-1.0	0.02	-1.0	0.02	mA
G31	111	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G32,G33	111	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G34	111	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G35	111	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G36	111	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G37	111	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G38	111	VIII	X	-3.0	0.1	-3.0	0.1	-3.0	0.1	mA
G39	111	VIII	X	-7.0	0.1	-7.0	0.1	-7.0	0.1	mA
G40-G42	111	VIII	X	-1.0	0.04	-1.0	0.04	-1.0	0.04	mA
G43,G44	111	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

-See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 3/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G45	Iil	VIII	X	-0.65	0.07	-0.65	0.07	-0.65	0.07	mA
G46	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G47	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G48	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G49	Iil	VIII	X	-3.2	0.09	-3.2	0.09	-3.2	0.09	mA
G50	Iil	VIII	X	-2.6	0.07	-2.6	0.07	-2.6	0.07	mA
G51	Iil	VIII	X	-4.8	0.16	-4.8	0.16	-4.8	0.16	mA
G52	Iil	VIII	X	-3.0	0.1	-3 0	0.1	-3.0	0 1	mA
G53, G54	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0 02	mA
G55	Iil	VIII	X	-2.4	0.08	-2.4	0.08	-2.4	0.08	mA
G56	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G57	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
H1-H31	Voh	IX	X	2 4	4.5	2.4	4.5	2 4	4 5	V
J1-J31	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1, K2	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K3, K4	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L2	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L3, L4	Vic	XII	S	-1 5	0.0	-1 5	0.0	-1.5	0 0	V
L5-L33	Vic	XII	S	-1 2	0.0	-1 2	0.0	-1.2	0 0	V
L34-L36	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L37-L50	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1, M2	PSFC	XIII	S	3.5	6.5	3.5	6.5	3.5	6.5	uF
N	Icc	XIV	X	550	1320	550	1320	550	1320	mA
P1, P2	Functional	XV	X	----------		See table XV				

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R2	C1/D	XVIII	Q	8	125	8	125	8	125	pF
R3-R6	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R7	C1/D	XVIII	Q	6	100	6	100	6	100	pF
R8	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R9	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R10	C1/D	XVIII	Q	8	100	8	100	8	100	pF
R11	C1/D	XVIII	Q	6	75	6	75	6	75	pF
R12,R13	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R14,R15	C1/D	XVIII	Q	6	75	6	75	6	75	pF
R16,R17	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R18	C1/D	XVIII	Q	4	50	4	50	4	50	pF
R19-R24	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R25	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R26	C1/D	XVIII	Q	6	75	6	75	6	75	pF
R27-R29	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R30,R31	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R32,R33	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R34	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R35	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R36	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R37	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R38	C1/D	XVIII	Q	10	125	10	125	10	125	pF
R39	C1/D	XVIII	Q	5	150	5	150	5	150	pF
R40	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R41	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R42	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R43-R45	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R46	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R47	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R48	C1/D	XVIII	Q	6	125	6	125	6	125	pF

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R49	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R50	C1/D	XVIII	Q	6	200	6	200	6	200	pF
R51	C1/D	XVIII	Q	2	150	2	150	2	150	pF
R52,R53	C1/D	XVIII	Q	2	50	2	50	2	50	pF
R54	C1/D	XVIII	Q	2	150	2	150	2	150	pF
R55,R56	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R57	C1/D	XVIII	Q	4	75	4	75	4	75	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

-See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/433-1 for class I and M28787/433-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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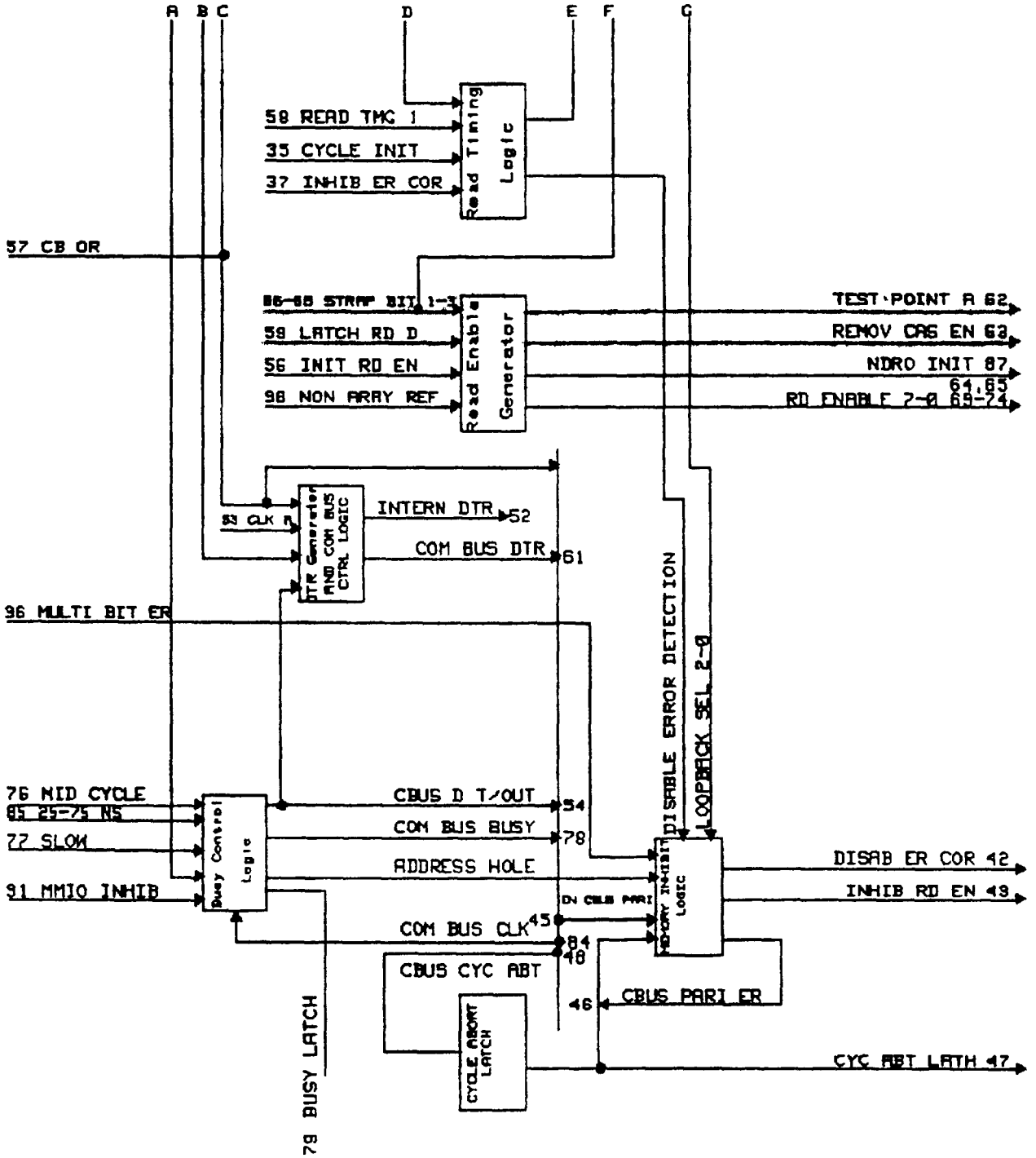


FIGURE 1. JEQ functional block diagram.

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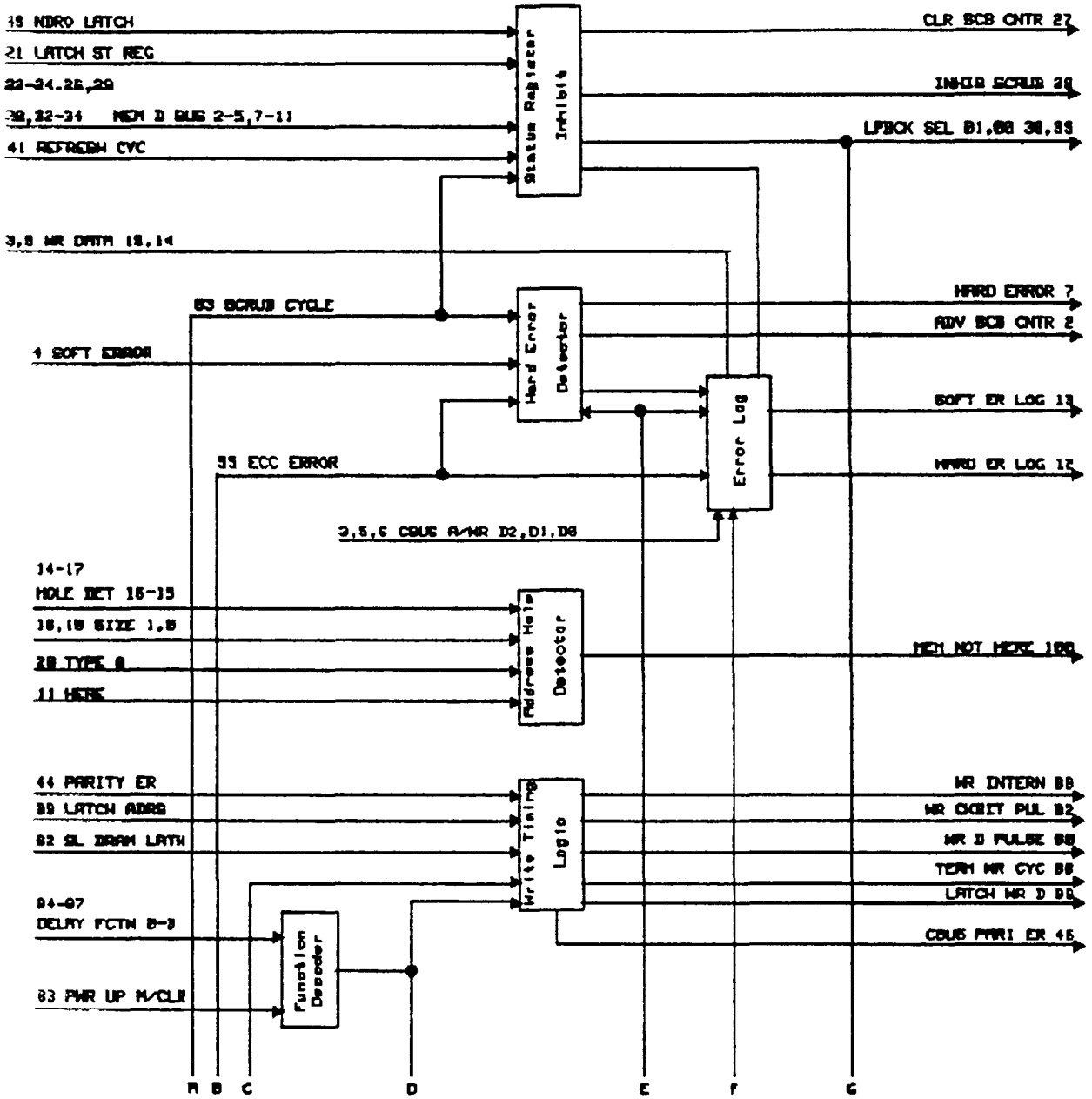


FIGURE 1. JEQ functional block diagram - Continued.

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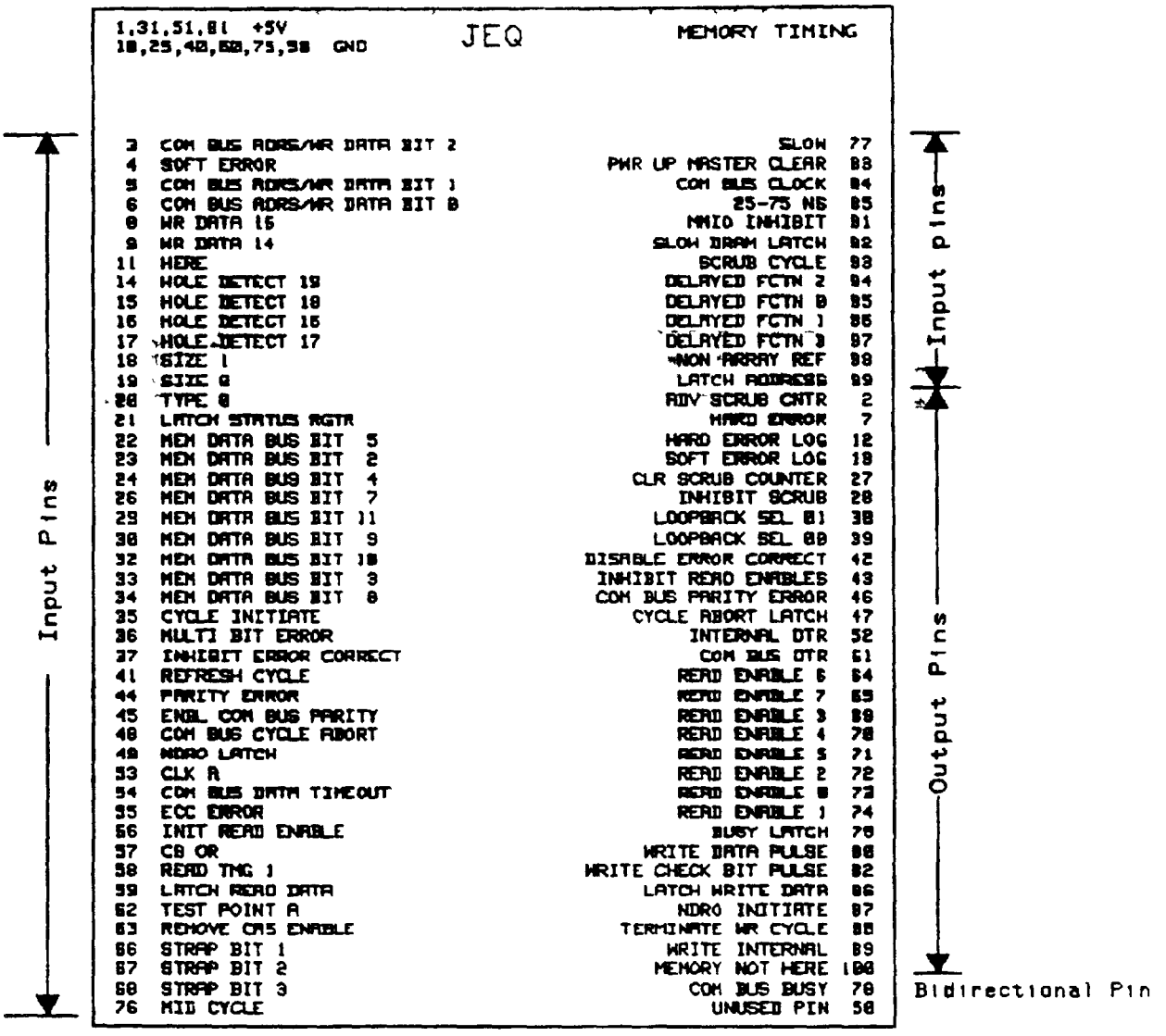


FIGURE 2. JEQ pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

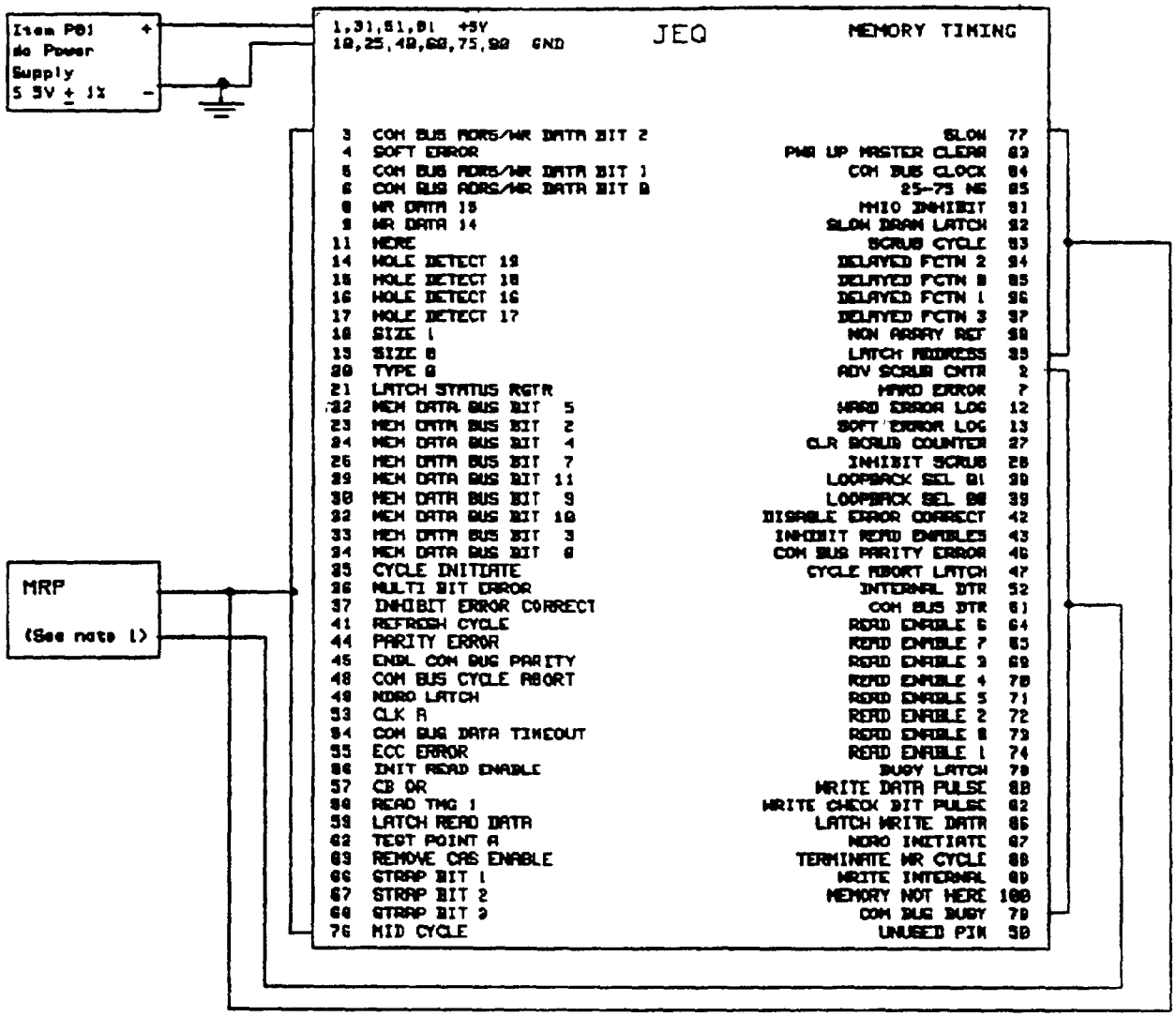
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. JEQ operating temperature and component temperature test circuit.

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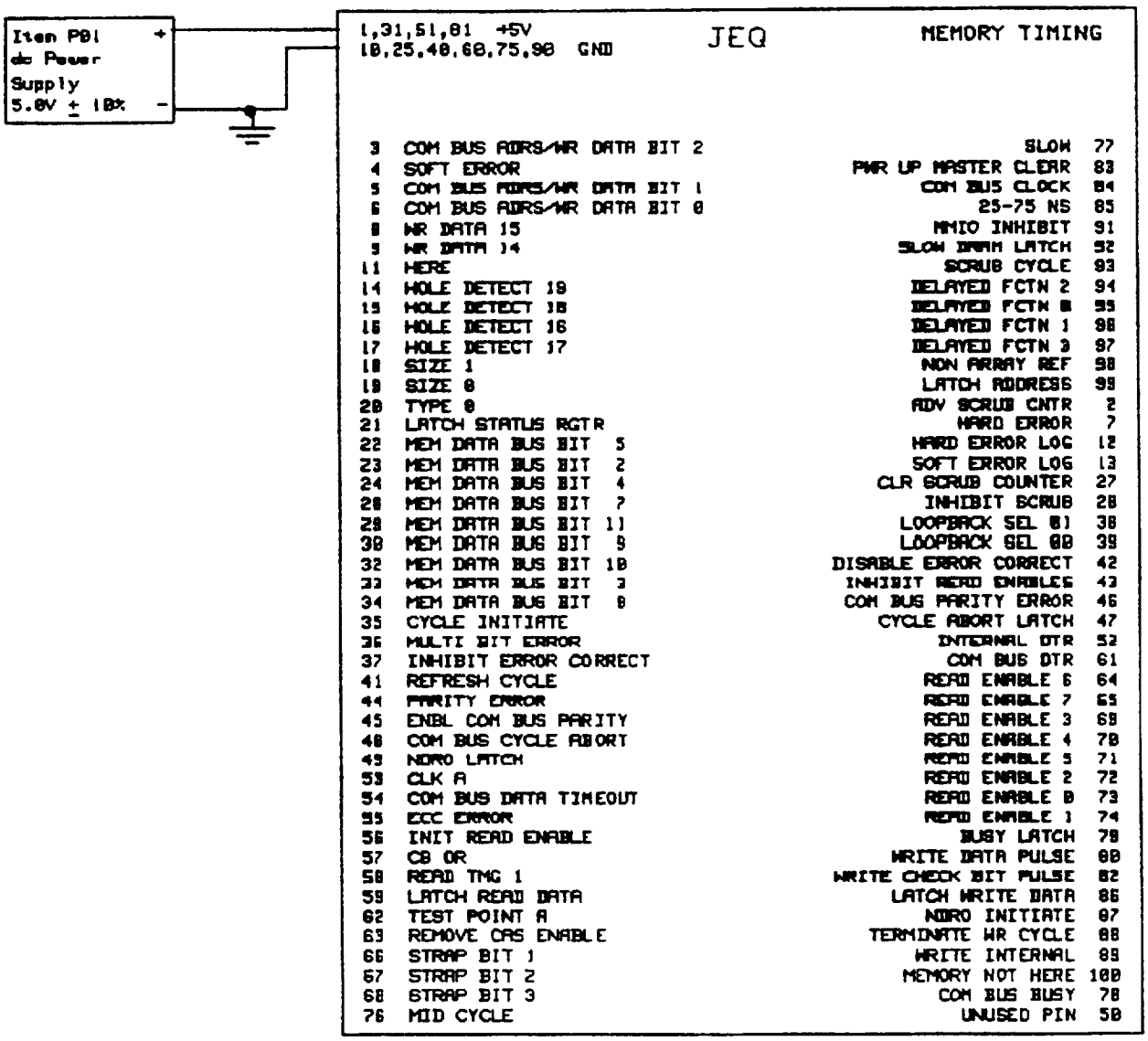


FIGURE 4. JEQ vibration test circuit.

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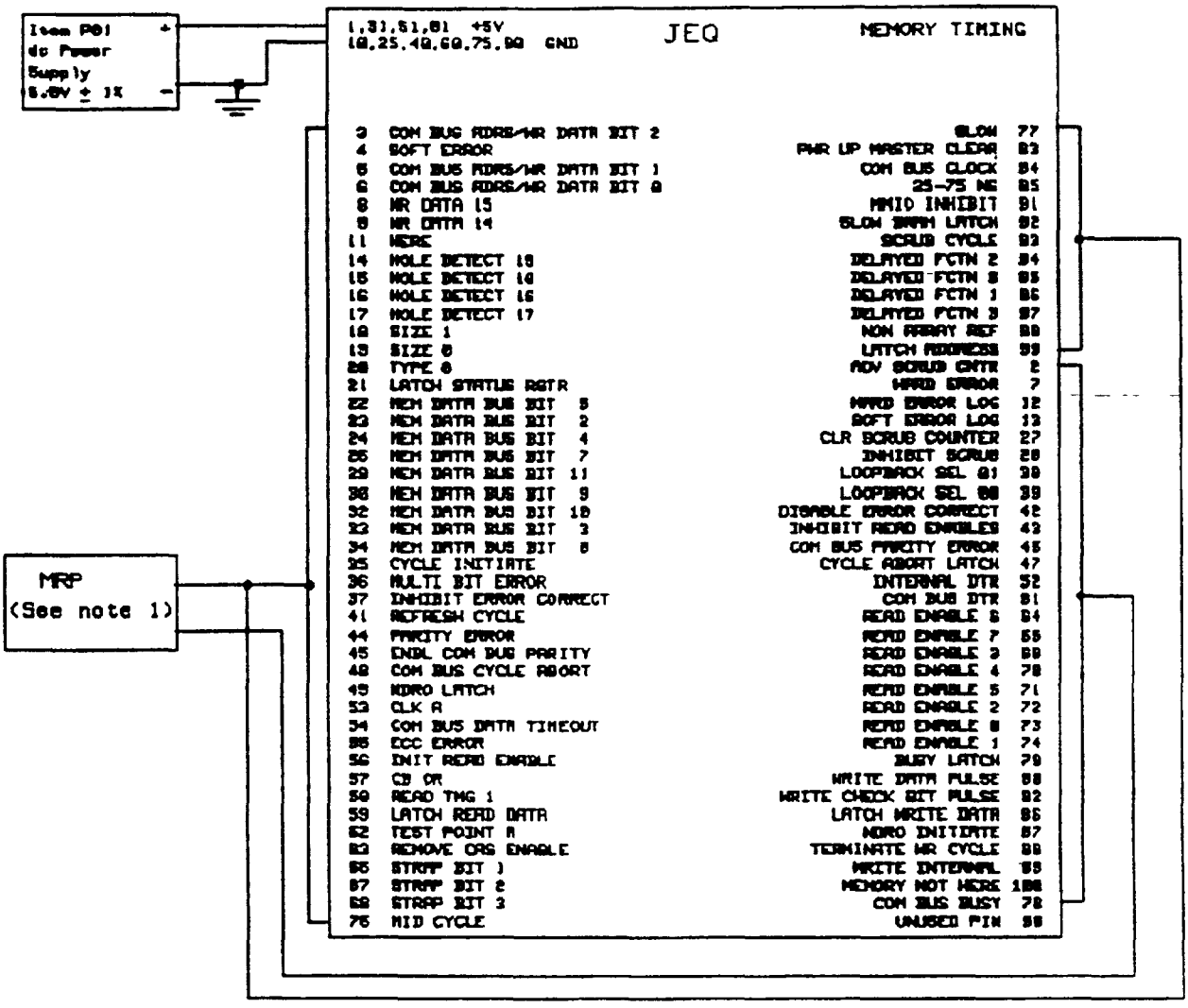
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. JEQ life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 7.26 W (max); 4.87 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CBUS A/WR D2	-1.5	0.4	2.4	-0.4	0.02	0.1	50
SOFT ERROR	-1.2	0.4	2.4	-2.4	0.08	0.4	125
CBUS A/WR D1	-1.5	0.4	2.4	-0.4	0.02	0.1	50
CBUS A/WR D0	-1.5	0.4	2.4	-0.4	0.02	0.1	50
WR DATA 15	-1.2	0.4	2.4	-0.6	0.02	0.1	50
WR DATA 14	-1.2	0.4	2.4	-0.6	0.02	0.1	50
HERE	NA	0.4	2.4	-28.3	-14.3	0.78	100
HOLE DET 19	-1.2	0.4	2.4	-0.6	0.02	0.1	50
HOLE DET 18	-1.2	0.4	2.4	-1.2	0.04	0.2	75
HOLE DET 16	-1.2	0.4	2.4	-2.4	0.08	0.4	100
HOLE DET 17	-1.2	0.4	2.4	-1.8	0.06	0.3	75
SIZE 1	NA	0.4	2.4	-27.7	-14.3	0.68	75
SIZE 0	NA	0.4	2.4	-27.7	-14.3	0.68	75
TYPE 0	NA	0.4	2.4	-28.3	-14.3	0.78	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
LATCH ST REG	-1.2	0.4	2.4	-1.8	0.06	0.3	75
MEM D BUS 5	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM D BUS 2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM D BUS 4	-1.2	0.4	2.4	-1.2	0.04	0.2	50
MEM D BUS 7	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM D BUS 11	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM D BUS 9	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM D BUS 10	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM D BUS 3	-1.2	0.4	2.4	-0.6	0.02	0.1	50
MEM D BUS 8	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CYCLE INIT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
MULTI BIT ER	-1.2	0.4	2.4	-1.8	0.06	0.3	75
INHIB ER COR	NA	0.4	2.4	-27.7	-14.3	0.68	75
REFRESH CYC	-1.2	0.4	2.4	-1.2	0.04	0.2	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
PARITY ER	NA	0.4	2.4	-28.3	-14.3	0.78	75
EN CBUS PARI	-1.2	0.4	2.4	-1.0	0.02	0.1	50
CBUS CYC ABT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
NDRO LATCH	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CLK A	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CBUS D T/OUT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ECC ERROR	-1.2	0.4	2.4	-1.2	0.04	0.2	75
INIT RD EN	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB OR	-1.2	0.4	2.4	-1.2	0.04	0.2	100
READ TMG 1	-1.2	0.4	2.4	-3.0	0.1	0.5	125
LATCH RD D	-1.2	0.4	2.4	-7.0	0.1	0.5	150
TEST POINT A	NA	NA	NA	NA	NA	NA	NA
REMOV CAS EN	NA	NA	NA	NA	NA	NA	NA
STRAP BIT 1	-1.5	0.4	2.4	-1.0	0.04	0.2	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
STRAP BIT 2	-1.5	0.4	2.4	-1.0	0.04	0.2	100
STRAP BIT 3	-1.5	0.4	2.4	-1.0	0.04	0.2	75
MID CYCLE	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SLOW	-1.2	0.4	2.4	-0.6	0.02	0.1	50
PWR UP M/CLR	-1.2	0.4	2.4	-0.6	0.02	0.1	50
COM BUS CLK	-1.2	0.4	2.4	-1.8	0.06	0.3	100
25-75 NS	-1.2	0.4	2.4	-1.2	0.04	0.2	75
MMIO INHIB	-1.2	0.4	2.4	-3.2	0.09	1.2	125
SL DRAM LATH	-1.2	0.4	2.4	-2.6	0.07	1.1	75
SCRUB CYCLE	-1.2	0.4	2.4	-4.8	0.16	0.8	200
DELAY FCTN 2	-1.2	0.4	2.4	-3.0	0.1	0.5	150
DELAY FCTN 0	-1.2	0.4	2.4	-0.6	0.02	0.1	50
DELAY FCTN 1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
DELAY FCTN 3	-1.2	0.4	2.4	-2.4	0.08	0.4	150

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
NON ARAY REF	-1.2	0.4	2.4	-1.8	0.06	0.3	125
LATCH ADRS	-1.2	0.4	2.4	-1.2	0.04	0.2	125
COM BUS BUSY	NA	0.4	2.4	-0.65	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
ADV SCB CNTR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
HARD ERROR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
HARD ER LOG	2.4	0.5	-0.4	6.0	NA	NA	-12.0	NA
SOFT ER LOG	2.4	0.5	-0.4	6.0	NA	NA	-12.0	NA
CLR SCB CNTR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
INHIB SCRUB	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LPBCK SEL 01	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LPBCK SEL 00	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DISAB ER COR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
INHIB RD EN	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CBUS PARI ER	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
CYC ABT LATH	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
INTERN DTR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
COM BUS DTR	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
RD ENABLE 6	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
RD ENABLE 7	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
RD ENABLE 3	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
RD ENABLE 4	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
RD ENABLE 5	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
RD ENABLE 2	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
RD ENABLE 0	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
RD ENABLE 1	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
BUSY LATCH	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
WR D PULSE	2.4	0.5	-0.4	6.0	NA	NA	-300.0	NA
WR CKBIT PUL	2.4	0.5	-0.4	6.0	NA	NA	-100.0	NA
LATCH WR D	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
NDRO INIT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
TERM WR CYC	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
WR INTERN	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MEM NOT HERE	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
COM BUS BUSY	2.4	0.5	-0.4	6.0	NA	NA	-100.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	LATCH ST REG to INHIB SCRUB	-	-	-	30	30	30	nS	11
2	tpd	LATCH ST REG to CLR SCB CNTR	-	-	-	19	19	19	nS	11
3	tpd	REFRESH CYC to LPBCK SEL 01	-	-	-	50	50	50	nS	11
4	tpd	LATCH ADRS to ABORT LATCH	-	-	-	30	30	30	nS	11
5	tpd	LATCH RD D to CBUS PARI ER	-	-	-	40	40	40	nS	11
6	tpd	EN CBUS PARI to CBUS PARI ER	-	-	-	18	18	18	nS	11
7	tpd	SIZE 1 to MEM NOT HERE	-	-	-	41	41	41	nS	11
7	tpd	SIZE 0 to MEM NOT HERE	-	-	-	41	41	41	nS	11
7	tpd	TYPE 0 to MEM NOT HERE	-	-	-	41	41	41	nS	11
8	tpd	HOLE DET 19 to MEM NOT HERE	-	-	-	25	25	25	nS	11
8	tpd	HOLE DET 18 to MEM NOT HERE	-	-	-	25	25	25	nS	11

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Check the source to verify that this is the current version before use.

6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	HOLE DET 16 to MEM NOT HERE	-	-	-	25	25	25	nS	11
8	tpd	HOLE DET 17 to MEM NOT HERE	-	-	-	25	25	25	nS	11
9	tpd	CLK A to INTERN DTR	-	-	-	16	16	16	nS	11
10	tpd	NON ARAY REF to RD ENABLE 6	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 7	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 3	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 4	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 5	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 2	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 0	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 1	-	-	-	25	25	25	nS	11

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	NON ARAY REF to NDRO INIT	-	-	-	38	38	38	nS	11
12	tpd	LATCH ADRS to LATCH WR D	-	-	-	42	42	42	nS	11
13	tpd	DELAY FCTN 2 to LATCH WR D	-	-	-	31	31	31	nS	11

6.4 Module function. The memory timing module contains common bus control circuits, the error log Random Access Memory (RAM), status register logic and read/write timing logic. Common bus control circuits decode delayed common bus function bits which define the operation requested. The decoded common bus function bits are gated together with internal memory timing signals to provide common bus handshaking responses from memory timing chain events. Status register logic provides inhibits to error correction and the error log (for a non-array reference, a Non-Destructive Read Out (NDRO) reference or an abort cycle), clears the scrub counter, flags parity errors and decodes the loopback select bits from the memory status register on the memory data module. The error detector receives hard and soft error detection bits and stores this information in a 16 location by 4-bit error log RAM until read by the error detection and correction logic on the memory data module. The read/write timing logic provides read and write enables to the memory array module, generates a write enable to the error log RAM to allow error data updating, and sends a disable to the memory control module which turns off the RAS to the memory array module. The read/write timing logic also contains that portion of the NDRO interface which disables the read enables to the memory array modules, generates an initialization signal to the bootstrap memory/Real Time Clock (RTC) module, and inhibits Error Detection/Correction (EDC) for an NDRO reference.

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6.4.1 Description of I/O pins.

Signal name	Pin number	I/O	Description
CBUS A/WR D2	3	I	COM BUS ADRS/WR DATA BIT 2
SOFT ERROR	4	I	SOFT ERROR
CBUS A/WR D1	5	I	COM BUS ADRS/WR DATA BIT 1
CBUS A/WR D0	6	I	COM BUS ADRS/WR DATA BIT 0
WR DATA 15	8	I	WR DATA 15
WR DATA 14	9	I	WR DATA 14
HERE	11	I	HERE
HOLE DET 19	14	I	HOLE DETECT 19
HOLE DET 18	15	I	HOLE DETECT 18
HOLE DET 16	16	I	HOLE DETECT 16
HOLE DET 17	17	I	HOLE DETECT 17
SIZE 1	18	I	SIZE 1
SIZE 0	19	I	SIZE 0
TYPE 0	20	I	TYPE 0
LATCH ST REG	21	I	LATCH STATUS RGTR
MEM D BUS 5	22	I	MEM DATA BUS BIT 5
MEM D BUS 2	23	I	MEM DATA BUS BIT 2
MEM D BUS 4	24	I	MEM DATA BUS BIT 4
MEM D BUS 7	26	I	MEM DATA BUS BIT 7
MEM D BUS 11	29	I	MEM DATA BUS BIT 11

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
MEM D BUS 9	30	I	MEM DATA BUS BIT 9
MEM D BUS 10	32	I	MEM DATA BUS BIT 10
MEM D BUS 3	33	I	MEM DATA BUS BIT 3
MEM D BUS 8	34	I	MEM DATA BUS BIT 8
CYCLE INIT	35	I	CYCLE INITIATE
MULTI BIT ER	36	I	MULTI BIT ERROR
INHIB ER COR	37	I	INHIBIT ERROR CORRECT
REFRESH CYC	41	I	REFRESH CYCLE
PARITY ER	44	I	PARITY ERROR
EN CBUS PARI	45	I	ENBL COM BUS PARITY
CBUS CYC ABT	48	I	COM BUS CYCLE ABORT
NDRO LATCH	49	I	NDRO LATCH
CLK A	53	I	CLK A
CBUS D T/OUT	54	I	COM BUS DATA TIMEOUT
ECC ERROR	55	I	ECC ERROR
INIT RD EN	56	I	INIT READ ENABLE
CB OR	57	I	CB OR
READ TMG 1	58	I	READ TMG 1
LATCH RD D	59	I	LATCH READ DATA
TEST POINT A	62	I	TEST POINT A

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
REMOV CAS EN	63	I	REMOVE CAS ENABLE
STRAP BIT 1	66	I	STRAP BIT 1
STRAP BIT 2	67	I	STRAP BIT 2
STRAP BIT 3	68	I	STRAP BIT 3
MID CYCLE	76	I	MID CYCLE
SLOW	77	I	SLOW
PWR UP M/CLR	83	I	PWR UP MASTER CLEAR
COM BUS CLK	84	I	COM BUS CLOCK
25-75 NS	85	I	25-75 NS
MMIO INHIB	91	I	MMIO INHIBIT
SL DRAM LATH	92	I	SLOW DRAM LATCH
SCRUB CYCLE	93	I	SCRUB CYCLE
DELAY FCTN 2	94	I	DELAYED FCTN 2
DELAY FCTN 0	95	I	DELAYED FCTN 0
DELAY FCTN 1	96	I	DELAYED FCTN 1
DELAY FCTN 3	97	I	DELAYED FCTN 3
NON ARAY REF	98	I	NON ARRAY REF
LATCH ADRS	99	I	LATCH ADDRESS
COM BUS BUSY	78	I/O	COM BUS BUSY
ADV SCB CNTR	2	O	ADV SCRUB CNTR

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
HARD ERROR	7	O	HARD ERROR
HARD ER LOG	12	O	HARD ERROR LOG
SOFT ER LOG	13	O	SOFT ERROR LOG
CLR SCB CNTR	27	O	CLR SCRUB COUNTER
INHIB SCRUB	28	O	INHIBIT SCRUB
LPBCK SEL 01	38	O	LOOPBACK SEL 01
LPBCK SEL 00	39	O	LOOPBACK SEL 00
DISAB ER COR	42	O	DISABLE ERROR CORRECT
INHIB RD EN	43	O	INHIBIT READ ENABLES
CBUS PARI ER	46	O	COM BUS PARITY ERROR
CYC ABT LATH	47	O	CYCLE ABORT LATCH
INTERN DTR	52	O	INTERNAL DTR
COM BUS DTR	61	O	COM BUS DTR
RD ENABLE 6	64	O	READ ENABLE 6
RD ENABLE 7	65	O	READ ENABLE 7
RD ENABLE 3	69	O	READ ENABLE 3
RD ENABLE 4	70	O	READ ENABLE 4
RD ENABLE 5	71	O	READ ENABLE 5
RD ENABLE 2	72	O	READ ENABLE 2
RD ENABLE 0	73	O	READ ENABLE 0

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6.4.1 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
RD ENABLE 1	74	O	READ ENABLE 1
BUSY LATCH	79	O	BUSY LATCH
WR D PULSE	80	O	WRITE DATA PULSE
WR CKBIT PUL	82	O	WRITE CHECK BIT PULSE
LATCH WR D	86	O	LATCH WRITE DATA
NDRO INIT	87	O	NDRO INITIATE
TERM WR CYC	88	O	TERMINATE WR CYCLE
WR INTERN	89	O	WRITE INTERNAL
MEM NOT HERE	100	O	MEMORY NOT HERE

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0070)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,50	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused pins	C	50	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1,A2
Isolation measurement	NA	± 500 KOhm	A1,A2

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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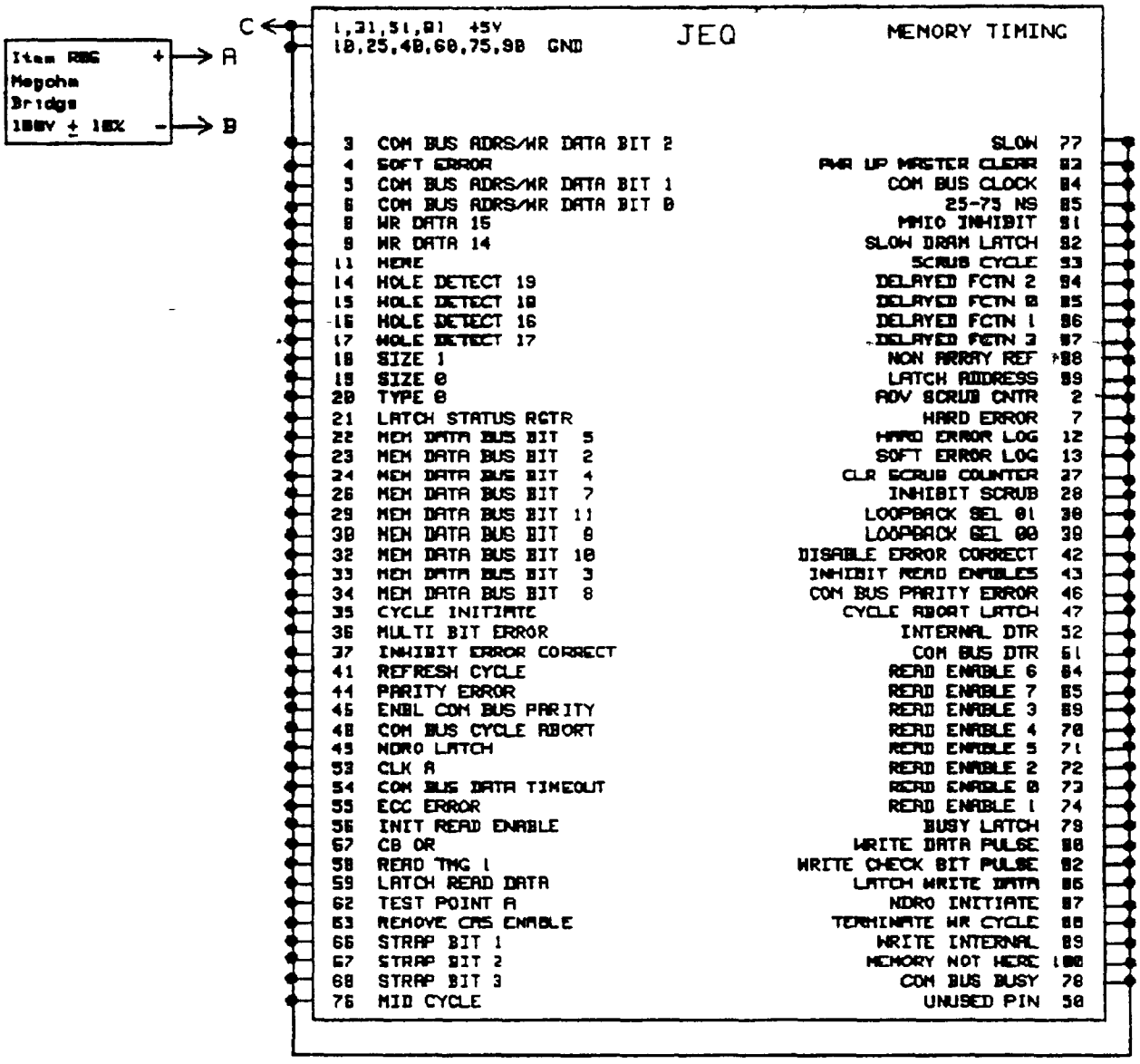


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

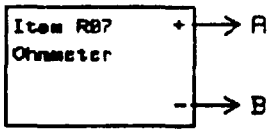
2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 +5V
10,25,40,60,75,90 GND

JEQ

MEMORY TIMING

3	COM BUS ADRS/WR DATA BIT 2	SLOW	77
4	SOFT ERROR	PWR UP MASTER CLEAR	83
5	COM BUS ADRS/WR DATA BIT 1	COM BUS CLOCK	84
6	COM BUS ADRS/WR DATA BIT 0	25-75 NS	85
8	WR DATA 15	MMIO INHIBIT	91
9	WR DATA 14	SLOW DRAM LATCH	92
11	HERE	SCRUB CYCLE	93
14	HOLE DETECT 19	DELAYED FCTN 2	94
15	HOLE DETECT 18	DELAYED FCTN 0	95
16	HOLE DETECT 16	DELAYED FCTN 1	96
17	HOLE DETECT 17	DELAYED FCTN 3	97
18	SIZE 1	NON-ARRAY REF	98
19	SIZE 0	LATCH ADDRESS	99
20	TYPE 0	ADV SCRUB CNTR	2
21	LATCH STATUS RGTR	HARD ERROR	7
22	MEM DATA BUS BIT 5	HARD ERROR LOG	12
23	MEM DATA BUS BIT 2	SOFT ERROR LOG	13
24	MEM DATA BUS BIT 4	CLR SCRUB COUNTER	27
26	MEM DATA BUS BIT 7	INHIBIT SCRUB	28
29	MEM DATA BUS BIT 11	LOOPBACK SEL B1	38
30	MEM DATA BUS BIT 9	LOOPBACK SEL B0	39
32	MEM DATA BUS BIT 10	DISABLE ERROR CORRECT	42
33	MEM DATA BUS BIT 3	INHIBIT READ ENABLES	43
34	MEM DATA BUS BIT 8	COM BUS PARITY ERROR	46
35	CYCLE INITIATE	CYCLE ABORT LATCH	47
36	MULTI BIT ERROR	INTERNAL DTR	52
37	INHIBIT ERROR CORRECT	COM BUS DTR	61
41	REFRESH CYCLE	READ ENABLE 6	64
44	PARITY ERROR	READ ENABLE 7	65
45	ENBL COM BUS PARITY	READ ENABLE 3	69
48	COM BUS CYCLE ABORT	READ ENABLE 4	70
49	NDRO LATCH	READ ENABLE 5	71
53	CLK A	READ ENABLE 2	72
54	COM BUS DATA TIMEOUT	READ ENABLE 0	73
55	ECC ERROR	READ ENABLE 1	74
56	INIT READ ENABLE	BUSY LATCH	79
57	CB OR	WRITE DATA PULSE	80
58	READ TMG 1	WRITE CHECK BIT PULSE	82
59	LATCH READ DATA	LATCH WRITE DATA	86
62	TEST POINT A	NDRO INITIATE	87
63	REMOVE CAS ENABLE	TERMINATE WR CYCLE	88
66	STRAP BIT 1	WRITE INTERNAL	89
67	STRAP BIT 2	MEMORY NOT HERE	100
68	STRAP BIT 3	COM BUS BUSY	78
76	MID CYCLE	UNUSED PIN	58

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.376 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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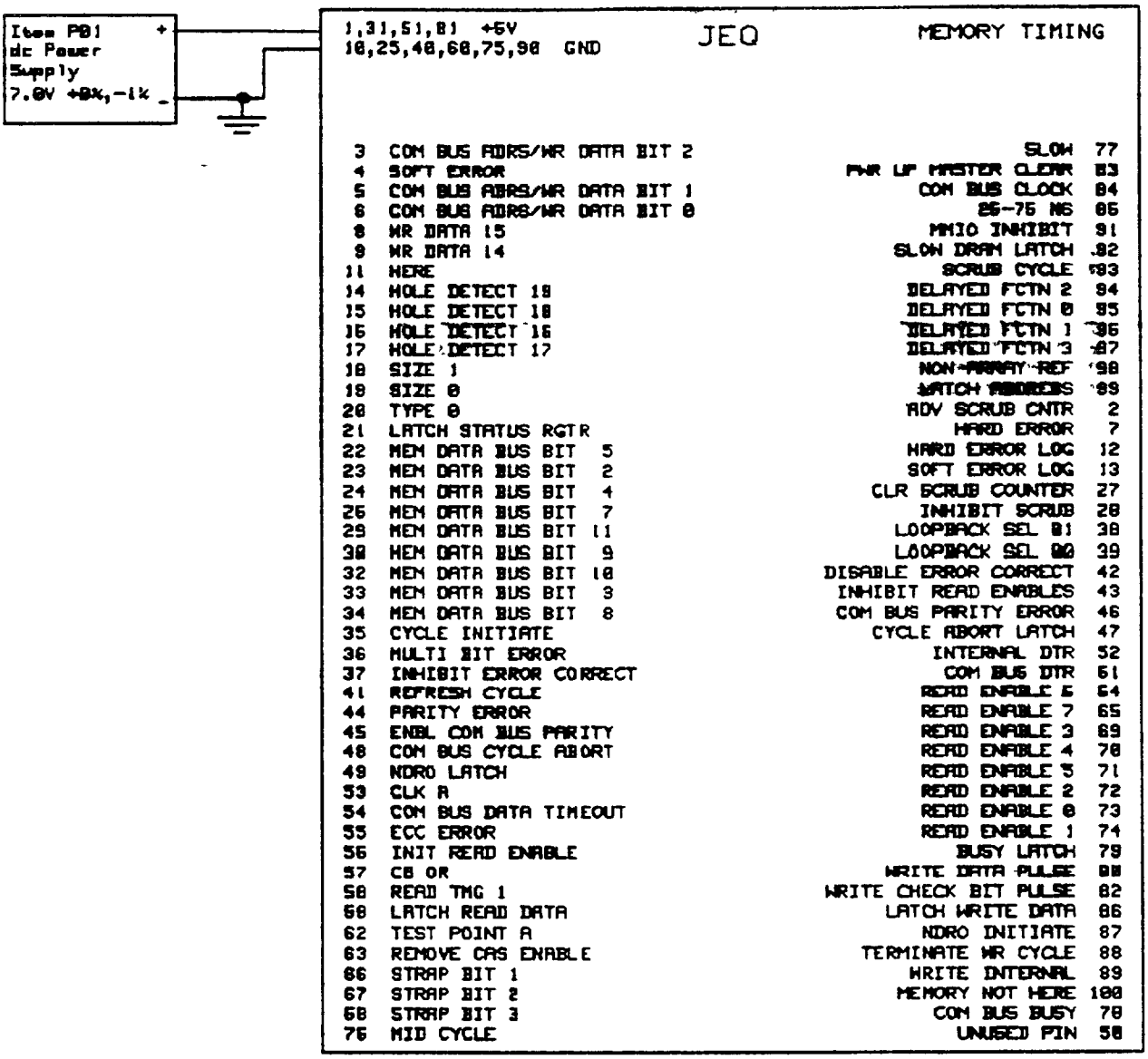


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	CBUS A/WR D2	3	5.5	0.0	0.1	mA	1-3
D2	SOFT ERROR	4	5.5	0.0	0.4	mA	1-3
D3	CBUS A/WR D1	5	5.5	0.0	0.1	mA	1-3
D4	CBUS A/WR D0	6					
D5	WR DATA 15	8					
D6	WR DATA 14	9	5.5	0.0	0.1	mA	1-3
D7	HERE	11	5.5	-0.58	0.78	mA	1-3
D8	HOLE DET 19	14	5.5	0.0	0.1	mA	1-3
D9	HOLE DET 18	15	5.5	0.0	0.2	mA	1-3
D10	HOLE DET 16	16	5.5	0.0	0.4	mA	1-3
D11	HOLE DET 17	17	5.5	0.0	0.3	mA	1-3
D12	SIZE 1	18	5.5	-0.58	0.68	mA	1-3
D13	SIZE 0	19	5.5	-0.58	0.68	mA	1-3
D14	TYPE 0	20	5.5	-0.58	0.78	mA	1-3
D15	LATCH ST REG	21	5.5	0.0	0.3	mA	1-3
D16	MEM D BUS 5	22	5.5	0.0	0.1	mA	1-3
D17	MEM D BUS 2	23	5.5	0.0	0.1	mA	1-3
D18	MEM D BUS 4	24	5.5	0.0	0.2	mA	1-3
D19	MEM D BUS 7	26	5.5	0.0	0.1	mA	1-3
D20	MEM D BUS 11	29	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	MEM D BUS 9	30	5.5	0.0	0.1	mA	1-3
D22	MEM D BUS 10	32					
D23	MEM D BUS 3	33					
D24	MEM D BUS 8	34					
D25	CYCLE INIT	35	5.5	0.0	0.1	mA	1-3
D26	MULTI BIT ER	36	5.5	0.0	0.3	mA	1-3
D27	INHIB ER COR	37	5.5	-0.58	0.68	mA	1-3
D28	REFRESH CYC	41	5.5	0.0	0.2	mA	1-3
D29	PARITY ER	44	5.5	-0.58	0.78	mA	1-3
D30	EN CBUS PARI	45	5.5	0.0	0.1	mA	1-3
D31	CBUS CYC ABT	48	5.5	0.0	0.1	mA	1-3
D32	NDRO LATCH	49	5.5	0.0	0.2	mA	1-3
D33	CLK A	53	5.5	0.0	0.2	mA	1-3
D34	CBUS D T/OUT	54	5.5	0.0	0.1	mA	1-3
D35	ECC ERROR	55	5.5	0.0	0.2	mA	1-3
D36	INIT RD EN	56	5.5	0.0	0.1	mA	1-3
D37	CB OR	57	5.5	0.0	0.2	mA	1-3
D38	READ TMG 1	58	5.5	0.0	0.5	mA	1-3
D39	LATCH RD D	59	5.5	0.0	0.5	mA	1-3
D40	STRAP BIT 1	66	5.5	0.0	0.2	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	STRAP BIT 2	67	5.5	0.0	0.2	mA	1-3
D42	STRAP BIT 3	68	5.5	0.0	0.2	mA	1-3
D43	MID CYCLE	76	5.5	0.0	0.1	mA	1-3
D44	SLOW	77					
D45	PWR UP M/CLR	83	5.5	0.0	0.1	mA	1-3
D46	COM BUS CLK	84	5.5	0.0	0.3	mA	1-3
D47	25-75 NS	85	5.5	0.0	0.2	mA	1-3
D48	MMIO INHIB	91	5.5	0.0	1.2	mA	1-3
D49	SL DRAM LATH	92	5.5	0.0	1.1	mA	1-3
D50	SCRUB CYCLE	93	5.5	0.0	0.8	mA	1-3
D51	DELAY FCTN 2	94	5.5	0.0	0.5	mA	1-3
D52	DELAY FCTN 0	95	5.5	0.0	0.1	mA	1-3
D53	DELAY FCTN 1	96	5.5	0.0	0.1	mA	1-3
D54	DELAY FCTN 3	97	5.5	0.0	0.4	mA	1-3
D55	NON ARAY REF	98	5.5	0.0	0.3	mA	1-3
D56	LATCH ADRS	99	5.5	0.0	0.2	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D56
Power supply current limit	1.98 A	NA	D1-D56
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D56
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D56
low, (VL)	0.0 V	± 100 mV	D1-D56
Input current measurement	NA	± 5 uA	D1
	NA	± 20 uA	D2
	NA	± 5 uA	D3-D6
	NA	± 68 uA	D7
	NA	± 5 uA	D8
	NA	± 10 uA	D9
	NA	± 20 uA	D10
	NA	± 15 uA	D11
	NA	± 63 uA	D12,D13
	NA	± 68 uA	D14
	NA	± 15 uA	D15
	NA	± 5 uA	D16,D17
	NA	± 10 uA	D18
	NA	± 5 uA	D19-D25
	NA	± 15 uA	D26
	NA	± 63 uA	D27
	NA	± 10 uA	D28
	NA	± 68 uA	D29
	NA	± 5 uA	D30,D31
	NA	± 10 uA	D32,D33
	NA	± 5 uA	D34

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 10 uA	D35
	NA	± 5 uA	D36
	NA	± 10 uA	D37
	NA	± 25 uA	D38,D39
	NA	± 10 uA	D40-D42
	NA	± 5 uA	D43-D45
	NA	± 15 uA	D46
	NA	± 10 uA	D47
	NA	± 60 uA	D48
	NA	± 55 uA	D49
	NA	± 40 uA	D50
	NA	± 25 uA	D51
	NA	± 5 uA	D52,D53
	NA	± 20 uA	D54
	NA	± 15 uA	D55
	NA	± 10 uA	D56

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	ADV SCB CNTR	2	0.0	-150.0	-60.0	mA	1-4
E2	HARD ERROR	7	0.0	-150.0	-60.0	mA	1-4
E3	HARD ER LOG	12	0.0	-90.0	-12.0	mA	1-4
E4	SOFT ER LOG	13	0.0	-90.0	-12.0	mA	1-4
E5	CLR SCB CNTR	27	0.0	-150.0	-60.0	mA	1-4
E6	INHIB SCRUB	28					
E7	LPBCK SEL 01	38					
E8	LPBCK SEL 00	39					
E9	DISAB ER COR	42					
E10	INHIB RD EN	43	0.0	-150.0	-60.0	mA	1-4
E11	CBUS PARI ER	46	0.0	-325.0	-100.0	mA	1-4
E12	CYC ABT LATH	47	0.0	-150.0	-60.0	mA	1-4
E13	INTERN DTR	52	0.0	-150.0	-60.0	mA	1-4
E14	COM BUS DTR	61	0.0	-325.0	-100.0	mA	1-4
E15	RD ENABLE 6	64	0.0	-100.0	-40.0	mA	1-4
E16	RD ENABLE 7	65					
E17	RD ENABLE 3	69					
E18	RD ENABLE 4	70					
E19	RD ENABLE 5	71					
E20	RD ENABLE 2	72	0.0	-100.0	-40.0	mA	1-4

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Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	RD ENABLE 0	73	0.0	-100.0	-40.0	mA	1-4
E22	RD ENABLE 1	74	0.0	-100.0	-40.0	mA	1-4
E23	COM BUS BUSY	78	0.0	-325.0	-100.0	mA	1-4
E24	BUSY LATCH	79	0.0	-150.0	-60.0	mA	1-4
E25	WR D PULSE	80	0.0	-975.0	-300.0	mA	1-4
E26	WR CKBIT PUL	82	0.0	-325.0	-100.0	mA	1-4
E27	LATCH WR D	86	0.0	-150.0	-60.0	mA	1-4
E28	NDRO INIT	87	0.0	-150.0	-60.0	mA	1-4
E29	TERM WR CYC	88	0.0	-100.0	-40.0	mA	1-4
E30	WR INTERN	89	0.0	-150.0	-60.0	mA	1-4
E31	MEM NOT HERE	100	0.0	-150.0	-60.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E31
Power supply current limit	1.98 A	NA	E1-E31
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E31

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E31
low, (VL)	0.0 V	± 100 mV	E1-E31
Output current measurement	NA	± 4.5 mA	E1,E2
	NA	± 3.9 mA	E3,E4
	NA	± 4.5 mA	E5-E10
	NA	± 11.2 mA	E11
	NA	± 4.5 mA	E12,E13
	NA	± 11.2 mA	E14
	NA	± 3.0 mA	E15-E22
	NA	± 11.2 mA	E23
	NA	± 4.5 mA	E24
	NA	± 33.7 mA	E25
	NA	± 11.2 mA	E26
	NA	± 4.5 mA	E27,E28
	NA	± 3.0 mA	E29
	NA	± 4.5 mA	E30,E31

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

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NOTES: - Continued.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	CBUS A/WR D2	3	2.4	-0.4	0.02	mA	1-3
F2	SOFT ERROR	4	2.4	-2.4	0.08	mA	1-3
F3	CBUS A/WR D1	5	2.4	-0.4	0.02	mA	1-3
F4	CBUS A/WR D0	6	2.4	-0.4	0.02	mA	1-3
F5	WR DATA 15	8	2.4	-0.6	0.02	mA	1-3
F6	WR DATA 14	9	2.4	-0.6	0.02	mA	1-3
F7	HERE	11	2.4	-28.3	-14.3	mA	1-3
F8	HOLE DET 19	14	2.4	-0.6	0.02	mA	1-3
F9	HOLE DET 18	15	2.4	-1.2	0.04	mA	1-3
F10	HOLE DET 16	16	2.4	-2.4	0.08	mA	1-3
F11	HOLE DET 17	17	2.4	-1.8	0.06	mA	1-3
F12	SIZE 1	18	2.4	-27.7	-14.3	mA	1-3
F13	SIZE 0	19	2.4	-27.7	-14.3	mA	1-3
F14	TYPE 0	20	2.4	-28.3	-14.3	mA	1-3
F15	LATCH ST REG	21	2.4	-1.8	0.06	mA	1-3
F16	MEM D BUS 5	22	2.4	-0.6	0.02	mA	1-3
F17	MEM D BUS 2	23	2.4	-0.6	0.02	mA	1-3
F18	MEM D BUS 4	24	2.4	-1.2	0.04	mA	1-3
F19	MEM D BUS 7	26	2.4	-0.6	0.02	mA	1-3
F20	MEM D BUS 11	29	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	MEM D BUS 9	30	2.4	-0.6	0.02	mA	1-3
F22	MEM D BUS 10	32					
F23	MEM D BUS 3	33					
F24	MEM D BUS 8	34					
F25	CYCLE INIT	35	2.4	-0.6	0.02	mA	1-3
F26	MULTI BIT ER	36	2.4	-1.8	0.06	mA	1-3
F27	INHIB ER COR	37	2.4	-27.7	-14.3	mA	1-3
F28	REFRESH CYC	41	2.4	-1.2	0.04	mA	1-3
F29	PARITY ER	44	2.4	-28.3	-14.3	mA	1-3
F30	EN CBUS PARI	45	2.4	-1.0	0.02	mA	1-3
F31	CBUS CYC ABT	48	2.4	-0.6	0.02	mA	1-3
F32	NDRO LATCH	49	2.4	-1.2	0.04	mA	1-3
F33	CLK A	53	2.4	-1.2	0.04	mA	1-3
F34	CBUS D T/OUT	54	2.4	-0.6	0.02	mA	1-3
F35	ECC ERROR	55	2.4	-1.2	0.04	mA	1-3
F36	INIT RD EN	56	2.4	-0.6	0.02	mA	1-3
F37	CB OR	57	2.4	-1.2	0.04	mA	1-3
F38	READ TMG 1	58	2.4	-3.0	0.1	mA	1-3
F39	LATCH RD D	59	2.4	-7.0	0.1	mA	1-3
F40	STRAP BIT 1	66	2.4	-1.0	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	STRAP BIT 2	67	2.4	-1.0	0.04	mA	1-3
F42	STRAP BIT 3	68	2.4	-1.0	0.04	mA	1-3
F43	MID CYCLE	76	2.4	-0.6	0.02	mA	1-3
F44	SLOW	77	2.4	-0.6	0.02	mA	1-3
F45	COM BUS BUSY	78	2.4	-0.65	0.07	mA	1-3
F46	PWR UP M/CLR	83	2.4	-0.6	0.02	mA	1-3
F47	COM BUS CLK	84	2.4	-1.8	0.06	mA	1-3
F48	25-75 NS	85	2.4	-1.2	0.04	mA	1-3
F49	MMIO INHIB	91	2.4	-3.2	0.09	mA	1-3
F50	SL DRAM LATH	92	2.4	-2.6	0.07	mA	1-3
F51	SCRUB CYCLE	93	2.4	-4.8	0.16	mA	1-3
F52	DELAY FCTN 2	94	2.4	-3.0	0.1	mA	1-3
F53	DELAY FCTN 0	95	2.4	-0.6	0.02	mA	1-3
F54	DELAY FCTN 1	96	2.4	-0.6	0.02	mA	1-3
F55	DELAY FCTN 3	97	2.4	-2.4	0.08	mA	1-3
F56	NON ARAY REF	98	2.4	-1.8	0.06	mA	1-3
F57	LATCH ADRS	99	2.4	-1.2	0.04	mA	1-3

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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F57
Power supply current limit	1.98 A	NA	F1-F57
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F57
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F57
low, (VL)	0.0 V	± 100 mV	F1-F57
Input current measurement	NA	± 21 uA	F1
	NA	± 124 uA	F2
	NA	± 21 uA	F3,F4
	NA	± 31 uA	F5,F6
	NA	± 700 uA	F7
	NA	± 31 uA	F8
	NA	± 62 uA	F9
	NA	± 124 uA	F10
	NA	± 93 uA	F11
	NA	± 670 uA	F12,F13
	NA	± 700 uA	F14
	NA	± 93 uA	F15
	NA	± 31 uA	F16,F17
	NA	± 62 uA	F18
	NA	± 31 uA	F19-F25
	NA	± 93 uA	F26
	NA	± 670 uA	F27
	NA	± 62 uA	F28
	NA	± 700 uA	F29
	NA	± 51 uA	F30
	NA	± 31 uA	F31

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 62 uA	F32,F33
	NA	± 31 uA	F34
	NA	± 62 uA	F35
	NA	± 31 uA	F36
	NA	± 62 uA	F37
	NA	± 155 uA	F38
	NA	± 360 uA	F39
	NA	± 52 uA	F40-F42
	NA	± 31 uA	F43,F44
	NA	± 36 uA	F45
	NA	± 31 uA	F46
	NA	± 93 uA	F47
	NA	± 62 uA	F48
	NA	± 164 uA	F49
	NA	± 133 uA	F50
	NA	± 248 uA	F51
	NA	± 155 uA	F52
	NA	± 31 uA	F53,F54
	NA	± 124 uA	F55
	NA	± 93 uA	F56
	NA	± 62 uA	F57

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	CBUS A/WR D2	3	0.4	-0.4	0.02	mA	1-3
G2	SOFT ERROR	4	0.4	-2.4	0.08	mA	1-3
G3	CBUS A/WR D1	5	0.4	-0.4	0.02	mA	1-3
G4	CBUS A/WR D0	6	0.4	-0.4	0.02	mA	1-3
G5	WR DATA 15	8	0.4	-0.6	0.02	mA	1-3
G6	WR DATA 14	9	0.4	-0.6	0.02	mA	1-3
G7	HERE	11	0.4	-28.3	-14.3	mA	1-3
G8	HOLE DET 19	14	0.4	-0.6	0.02	mA	1-3
G9	HOLE DET 18	15	0.4	-1.2	0.04	mA	1-3
G10	HOLE DET 16	16	0.4	-2.4	0.08	mA	1-3
G11	HOLE DET 17	17	0.4	-1.8	0.06	mA	1-3
G12	SIZE 1	18	0.4	-27.7	-14.3	mA	1-3
G13	SIZE 0	19	0.4	-27.7	-14.3	mA	1-3
G14	TYPE 0	20	0.4	-28.3	-14.3	mA	1-3
G15	LATCH ST REG	21	0.4	-1.8	0.06	mA	1-3
G16	MEM D BUS 5	22	0.4	-0.6	0.02	mA	1-3
G17	MEM D BUS 2	23	0.4	-0.6	0.02	mA	1-3
G18	MEM D BUS 4	24	0.4	-1.2	0.04	mA	1-3
G19	MEM D BUS 7	26	0.4	-0.6	0.02	mA	1-3
G20	MEM D BUS 11	29	0.4	-0.6	0.02	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	MEM D BUS 9	30	0.4	-0.6	0.02	mA	1-3
G22	MEM D BUS 10	32					
G23	MEM D BUS 3	33					
G24	MEM D BUS 8	34					
G25	CYCLE INIT	35	0.4	-0.6	0.02	mA	1-3
G26	MULTI BIT ER	36	0.4	-1.8	0.06	mA	1-3
G27	INHIB ER COR	37	0.4	-27.7	-14.3	mA	1-3
G28	REFRESH CYC	41	0.4	-1.2	0.04	mA	1-3
G29	PARITY ER	44	0.4	-28.3	-14.3	mA	1-3
G30	EN CBUS PARI	45	0.4	-1.0	0.02	mA	1-3
G31	CBUS CYC ABT	48	0.4	-0.6	0.02	mA	1-3
G32	NDRO LATCH	49	0.4	-1.2	0.04	mA	1-3
G33	CLK A	53	0.4	-1.2	0.04	mA	1-3
G34	CBUS D T/OUT	54	0.4	-0.6	0.02	mA	1-3
G35	ECC ERROR	55	0.4	-1.2	0.04	mA	1-3
G36	INIT RD EN	56	0.4	-0.6	0.02	mA	1-3
G37	CB OR	57	0.4	-1.2	0.04	mA	1-3
G38	READ TMG 1	58	0.4	-3.0	0.1	mA	1-3
G39	LATCH RD D	59	0.4	-7.0	0.1	mA	1-3
G40	STRAP BIT 1	66	0.4	-1.0	0.04	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	STRAP BIT 2	67	0.4	-1.0	0.04	mA	1-3
G42	STRAP BIT 3	68	0.4	-1.0	0.04	mA	1-3
G43	MID CYCLE	76	0.4	-0.6	0.02	mA	1-3
G44	SLOW	77	0.4	-0.6	0.02	mA	1-3
G45	COM BUS BUSY	78	0.4	-0.65	0.07	mA	1-3
G46	PWR UP M/CLR	83	0.4	-0.6	0.02	mA	1-3
G47	COM BUS CLK	84	0.4	-1.8	0.06	mA	1-3
G48	25-75 NS	85	0.4	-1.2	0.04	mA	1-3
G49	MMIO INHIB	91	0.4	-3.2	0.09	mA	1-3
G50	SL DRAM LATH	92	0.4	-2.6	0.07	mA	1-3
G51	SCRUB CYCLE	93	0.4	-4.8	0.16	mA	1-3
G52	DELAY FCTN 2	94	0.4	-3.0	0.1	mA	1-3
G53	DELAY FCTN 0	95	0.4	-0.6	0.02	mA	1-3
G54	DELAY FCTN 1	96	0.4	-0.6	0.02	mA	1-3
G55	DELAY FCTN 3	97	0.4	-2.4	0.08	mA	1-3
G56	NON ARAY REF	98	0.4	-1.8	0.06	mA	1-3
G57	LATCH ADRS	99	0.4	-1.2	0.04	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G57
Power supply current limit	1.98 A	NA	G1-G57
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G57
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G57 G1-G57
Input current measurement	NA	± 21 uA	G1
	NA	± 124 uA	G2
	NA	± 21 uA	G3,G4
	NA	± 31 uA	G5,G6
	NA	± 700 uA	G7
	NA	± 31 uA	G8
	NA	± 62 uA	G9
	NA	± 124 uA	G10
	NA	± 93 uA	G11
	NA	± 670 uA	G12,G13
	NA	± 700 uA	G14
	NA	± 93 uA	G15
	NA	± 31 uA	G16,G17
	NA	± 62 uA	G18
	NA	± 31 uA	G19-G25
	NA	± 93 uA	G26
	NA	± 670 uA	G27
	NA	± 62 uA	G28
	NA	± 700 uA	G29
	NA	± 51 uA	G30
	NA	± 31 uA	G31
	NA	± 62 uA	G32,G33

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 31 uA	G34
	NA	± 62 uA	G35
	NA	± 31 uA	G36
	NA	± 62 uA	G37
	NA	± 155 uA	G38
	NA	± 360 uA	G39
	NA	± 52 uA	G40-G42
	NA	± 31 uA	G43,G44
	NA	± 36 uA	G45
	NA	± 31 uA	G46
	NA	± 93 uA	G47
	NA	± 62 uA	G48
	NA	± 164 uA	G49
	NA	± 133 uA	G50
	NA	± 248 uA	G51
	NA	± 155 uA	G52
	NA	± 31 uA	G53,G54
	NA	± 124 uA	G55
	NA	± 93 uA	G56
	NA	± 62 uA	G57

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	ADV SCB CNTR	2	-0.4	2.4	4.5	V	1-4
H2	HARD ERROR	7					
H3	HARD ER LOG	12					
H4	SOFT ER LOG	13					
H5	CLR SCB CNTR	27					
H6	INHIB SCRUB	28					
H7	LPBCK SEL 01	38					
H8	LPBCK SEL 00	39					
H9	DISAB ER COR	42					
H10	INHIB RD EN	43					
H11	CBUS PARI ER	46					
H12	CYC ABT LATH	47					
H13	INTERN DTR	52					
H14	COM BUS DTR	61					
H15	RD ENABLE 6	64					
H16	RD ENABLE 7	65					
H17	RD ENABLE 3	69					
H18	RD ENABLE 4	70					
H19	RD ENABLE 5	71					
H20	RD ENABLE 2	72	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	RD ENABLE 0	73	-0.4	2.4	4.5	V	1-4
H22	RD ENABLE 1	74					
H23	COM BUS BUSY	78					
H24	BUSY LATCH	79					
H25	WR D PULSE	80					
H26	WR CKBIT PUL	82					
H27	LATCH WR D	86					
H28	NDRO INIT	87					
H29	TERM WR CYC	88					
H30	WR INTERN	89					
H31	MEM NOT HERE	100	-0.4	2.4	4.5	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H31
Power supply current limit	1.98 A	NA	H1-H31
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H31

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H31 H1-H31
Output voltage measurement	NA	± 105 mV	H1-H31

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

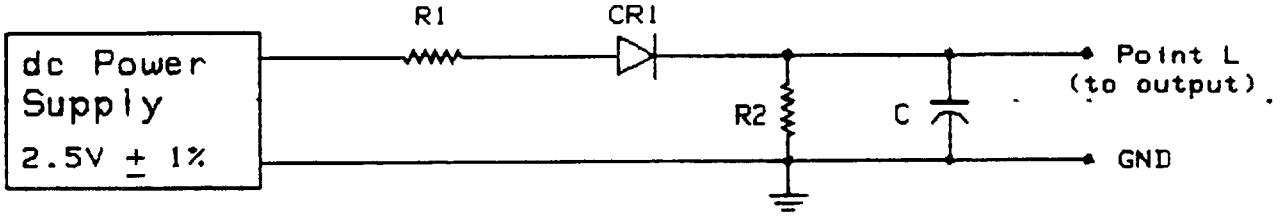
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

    (a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

    (b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

    (c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

    (d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,7,12,13,27,28, 38,39,42,43,46, 47,52,61,64,65, 69,70,71,72,73, 74,78,79,80,82, 86,87,88,89,100	6200	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	ADV SCB CNTR	2	6.0	0.0	0.5	V	1-4
J2	HARD ERROR	7					
J3	HARD ER LOG	12					
J4	SOFT ER LOG	13					
J5	CLR SCB CNTR	27					
J6	INHIB SCRUB	28					
J7	LPBCK SEL 01	38					
J8	LPBCK SEL 00	39					
J9	DISAB ER COR	42					
J10	INHIB RD EN	43					
J11	CBUS PARI ER	46					
J12	CYC ABT LATH	47					
J13	INTERN DTR	52					
J14	COM BUS DTR	61					
J15	RD ENABLE 6	64					
J16	RD ENABLE 7	65					
J17	RD ENABLE 3	69					
J18	RD ENABLE 4	70					
J19	RD ENABLE 5	71					
J20	RD ENABLE 2	72	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	RD ENABLE 0	73	6.0	0.0	0.5	V	1-4
J22	RD ENABLE 1	74					
J23	COM BUS BUSY	78					
J24	BUSY LATCH	79					
J25	WR D PULSE	80					
J26	WR CKBIT PUL	82					
J27	LATCH WR D	86					
J28	NDRO INIT	87					
J29	TERM WR CYC	88					
J30	WR INTERN	89					
J31	MEM NOT HERE	100	6.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J31
Power supply current limit	1.98 A	NA	J1-J31
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J31

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J31 J1-J31
Output voltage measurement	NA	± 25 mV	J1-J31

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	CBUS PARI ER	46	2.4	-0.05	0.05	mA	1-3
K2	COM BUS DTR	61	2.4	-0.05	0.05	mA	1-3
Low level							
K3	CBUS PARI ER	46	0.4	-0.05	0.05	mA	1-3
K4	COM BUS DTR	61	0.4	-0.05	0.05	mA	1-3

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K4
Power supply current limit	1.98 A	NA	K1-K4
Forcing voltage, (Voh) (Vol)	2.4 V 0.4 V	± 50 mV ± 50 mV	K1,K2 K3,K4
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1-K4 K1-K4
Output current measurement	NA	± 5 uA	K1-K4

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	CBUS A/WR D2	3	-18.0	-1.5	0.0	V	1-3
L2	SOFT ERROR	4	-18.0	-1.2	0.0	V	1-3
L3	CBUS A/WR D1	5	-18.0	-1.5	0.0	V	1-3
L4	CBUS A/WR D0	6	-18.0	-1.5	0.0	V	1-3
L5	WR DATA 15	8	-18.0	-1.2	0.0	V	1-3
L6	WR DATA 14	9					
L7	HOLE DET 19	14					
L8	HOLE DET 18	15					
L9	HOLE DET 16	16					
L10	HOLE DET 17	17					
L11	LATCH ST REG	21					
L12	MEM D BUS 5	22					
L13	MEM D BUS 2	23					
L14	MEM D BUS 4	24					
L15	MEM D BUS 7	26					
L16	MEM D BUS 11	29					
L17	MEM D BUS 9	30					
L18	MEM D BUS 10	32					
L19	MEM D BUS 3	33					
L20	MEM D BUS 8	34	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	CYCLE INIT	35	-18.0	-1.2	0.0	V	1-3
L22	MULTI BIT ER	36					
L23	REFRESH CYC	41					
L24	EN CBUS PARI	45					
L25	CBUS CYC ABT	48					
L26	NDRO LATCH	49					
L27	CLK A	53					
L28	CBUS D T/OUT	54					
L29	ECC ERROR	55					
L30	INIT RD EN	56					
L31	CB OR	57					
L32	READ TMG 1	58					
L33	LATCH RD D	59	-18.0	-1.2	0.0	V	1-3
L34	STRAP BIT 1	66	-18.0	-1.5	0.0	V	1-3
L35	STRAP BIT 2	67					
L36	STRAP BIT 3	68	-18.0	-1.5	0.0	V	1-3
L37	MID CYCLE	76	-18.0	-1.2	0.0	V	1-3
L38	SLOW	77					
L39	PWR UP M/CLR	83					
L40	COM BUS CLK	84	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L41	25-75 NS	85	-18.0	-1.2	0.0	V	1-3
L42	MMIO INHIB	91					
L43	SL DRAM LATH	92					
L44	SCRUB CYCLE	93					
L45	DELAY FCTN 2	94					
L46	DELAY FCTN 0	95					
L47	DELAY FCTN 1	96					
L48	DELAY FCTN 3	97					
L49	NON ARAY REF	98					
L50	LATCH ADRS	99	-18.0	-1.2	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L50
Power supply current limit	NA	NA	L1-L50
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L50

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L50 L1-L50
Input voltage measurement	NA NA NA NA NA NA	± 75 mV ± 60 mV ± 75 mV ± 60 mV ± 75 mV ± 60 mV	L1 L2 L3,L4 L5-L33 L34-L36 L37-L50

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	3.5	6.5	uF	1-4
M2	PSFC	51	3.5	6.5	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

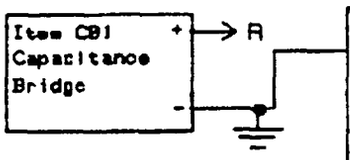
Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA	± 150 nF	M1,M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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1,31,51,81 +5V
10,25,40,60,75,90 GND

JEQ

MEMORY TIMING

3	COM BUS ADRS/WR DATA BIT 2	SLOW	77
4	SOFT ERROR	PWR UP MASTER CLEAR	83
5	COM BUS ADRS/WR DATA BIT 1	COM BUS CLOCK	84
6	COM BUS ADRS/WR DATA BIT 0	25-75 NS	85
8	WR DATA 15	MMIO INHIBIT	91
9	WR DATA 14	SLOW DRAM LATCH	92
11	HERE	SCRUB CYCLE	93
14	HOLE DETECT 18	DELAYED FCTN 2	94
15	HOLE DETECT 18	DELAYED FCTN 0	95
16	HOLE DETECT 16	DELAYED FCTN 1	96
17	HOLE DETECT 17	DELAYED FCTN 3	97
18	SIZE 1	NON ARRAY REF	98
19	SIZE 0	LATCH ADDRESS	99
20	TYPE 0	ADV SCRUB CNTR	2
21	LATCH STATUS RGTR	HARD ERROR	7
22	MEM DATA BUS BIT 5	HARD ERROR LOG	12
23	MEM DATA BUS BIT 2	SOFT ERROR LOG	13
24	MEM DATA BUS BIT 4	CLR SCRUB COUNTER	27
26	MEM DATA BUS BIT 7	INHIBIT SCRUB	28
29	MEM DATA BUS BIT 11	LOOPBACK SEL 01	38
30	MEM DATA BUS BIT 9	LOOPBACK SEL 00	39
32	MEM DATA BUS BIT 10	DISABLE ERROR CORRECT	42
33	MEM DATA BUS BIT 3	INHIBIT READ ENABLES	43
34	MEM DATA BUS BIT 8	COM BUS PARITY ERROR	46
35	CYCLE INITIATE	CYCLE ABORT LATCH	47
36	MULTI BIT ERROR	INTERNAL DTR	52
37	INHIBIT ERROR CORRECT	COM BUS DTR	61
41	REFRESH CYCLE	READ ENABLE 6	64
44	PARITY ERROR	READ ENABLE 7	65
45	ENBL COM BUS PARITY	READ ENABLE 3	69
48	COM BUS CYCLE ABORT	READ ENABLE 4	70
49	NDRO LATCH	READ ENABLE 5	71
53	CLK A	READ ENABLE 2	72
54	COM BUS DATA TIMEOUT	READ ENABLE 0	73
55	ECC ERROR	READ ENABLE 1	74
56	INIT READ ENABLE	BUSY LATCH	79
57	CB OR	WRITE DATA PULSE	80
58	READ TMG 1	WRITE CHECK BIT PULSE	82
59	LATCH READ DATA	LATCH WRITE DATA	86
62	TEST POINT A	NDRO INITIATE	87
63	REMOVE CAS ENABLE	TERMINATE WR CYCLE	88
66	STRAP BIT 1	WRITE INTERNAL	89
67	STRAP BIT 2	MEMORY NOT HERE	100
68	STRAP BIT 3	COM BUS BUSY	78
76	MID CYCLE	UNUSED PIN	50

FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	ICC	1,31, 51,81	550	1320	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.98 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 38.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.98 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level:	$2.4\ V \leq Voh \leq Vcc$
Low level:	$0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.98 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

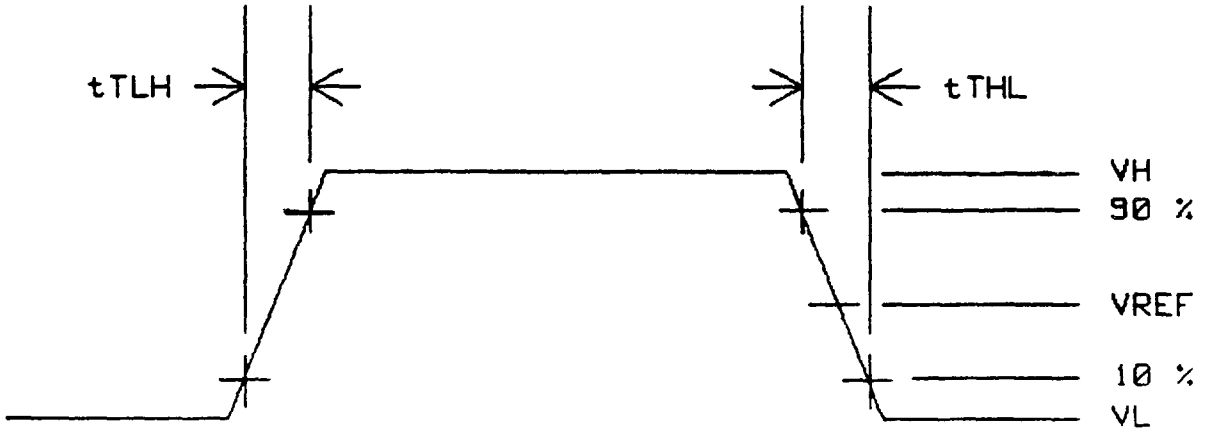
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	LATCH ST REG to INHIB SCRUB	-	-	-	30	30	30	nS	11
2	tpd	LATCH ST REG to CLR SCB CNTR	-	-	-	19	19	19	nS	11
3	tpd	REFRESH CYC to LPBCK SEL 01	-	-	-	50	50	50	nS	11
4	tpd	LATCH ADRS to ABORT LATCH	-	-	-	30	30	30	nS	11
5	tpd	LATCH RD D to CBUS PARI ER	-	-	-	40	40	40	nS	11
6	tpd	EN CBUS PARI to CBUS PARI ER	-	-	-	18	18	18	nS	11
7	tpd	SIZE 1 to MEM NOT HERE	-	-	-	41	41	41	nS	11
7	tpd	SIZE 0 to MEM NOT HERE	-	-	-	41	41	41	nS	11
7	tpd	TYPE 0 to MEM NOT HERE	-	-	-	41	41	41	nS	11
8	tpd	HOLE DET 19 to MEM NOT HERE	-	-	-	25	25	25	nS	11
8	tpd	HOLE DET 18 to MEM NOT HERE	-	-	-	25	25	25	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	HOLE DET 16 to MEM NOT HERE	-	-	-	25	25	25	nS	11
8	tpd	HOLE DET 17 to MEM NOT HERE	-	-	-	25	25	25	nS	11
9	tpd	CLK A to INTERN DTR	-	-	-	16	16	16	nS	11
10	tpd	NON ARAY REF to RD ENABLE 6	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 7	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 3	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 4	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 5	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 2	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 0	-	-	-	25	25	25	nS	11
10	tpd	NON ARAY REF to RD ENABLE 1	-	-	-	25	25	25	nS	11

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
11	tpd	NON ARAY REF to NDRO INIT	-	-	-	38	38	.38	nS	11
12	tpd	LATCH ADRS to LATCH WR D	-	-	-	42	42	42	nS	11
13	tpd	DELAY FCTN 2 to LATCH WR D	-	-	-	31	31	31	nS	11

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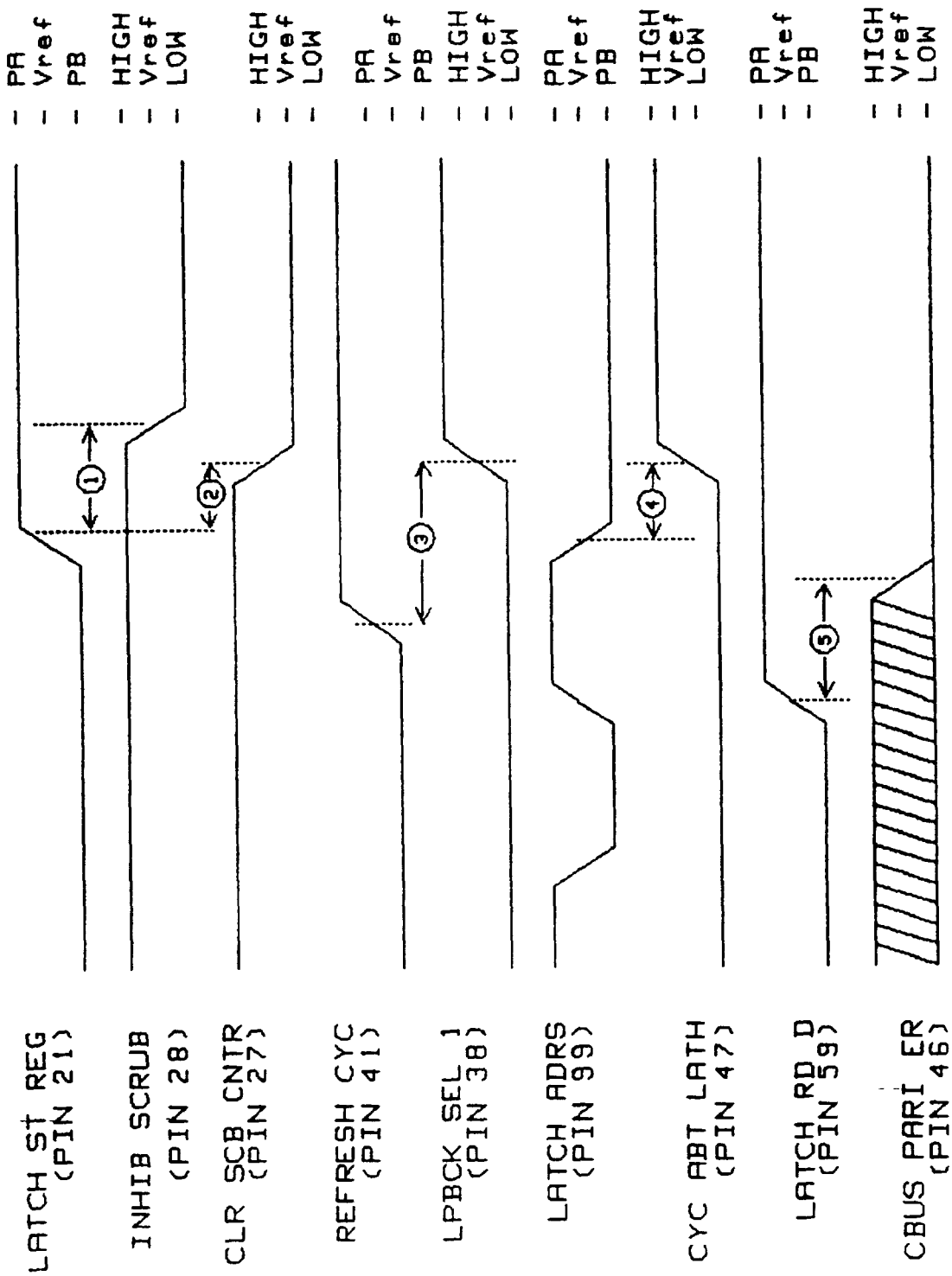


FIGURE 11a. Dynamic waveforms for table XVII.

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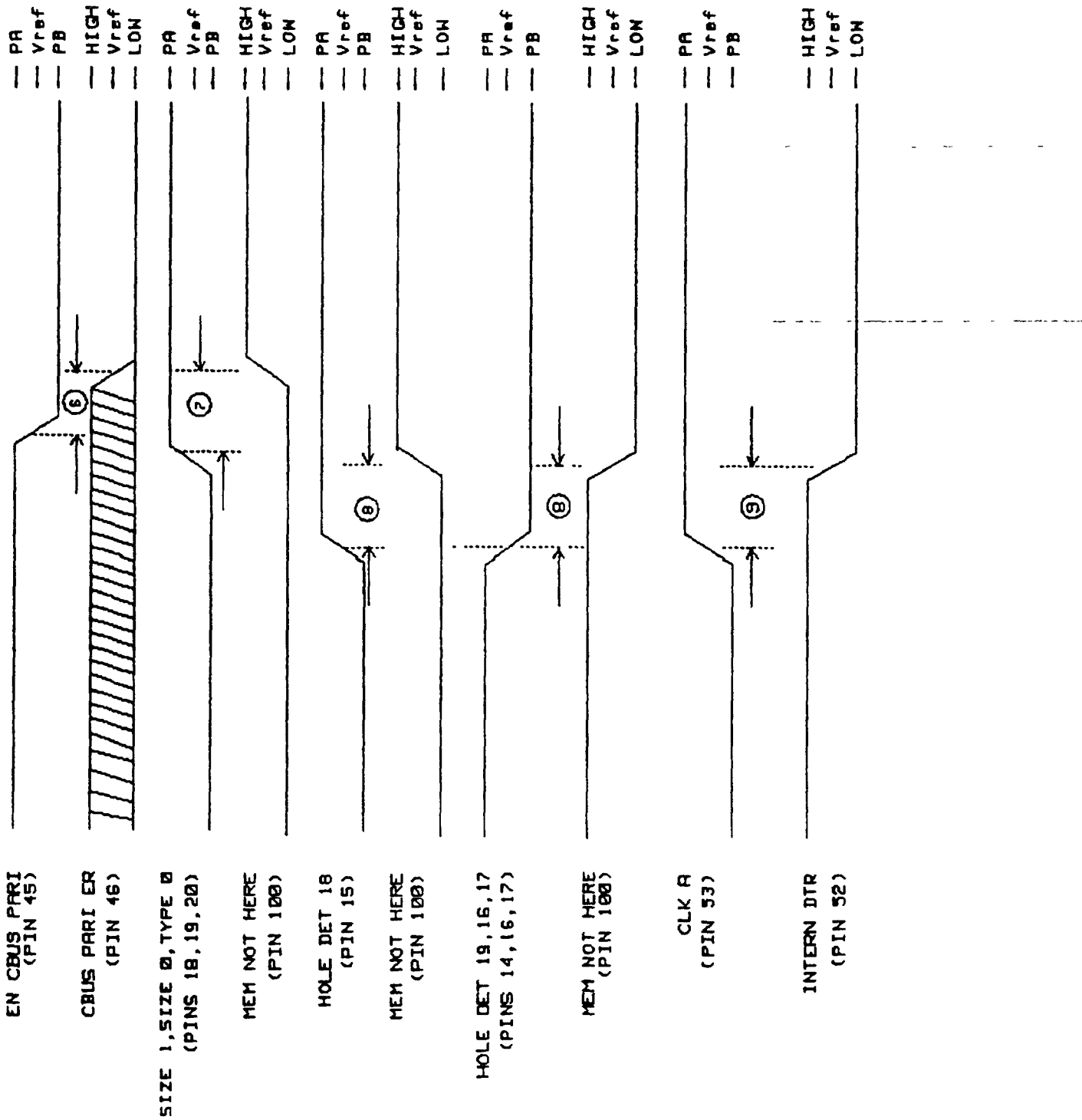


FIGURE 11b. Dynamic waveforms for table XVII.

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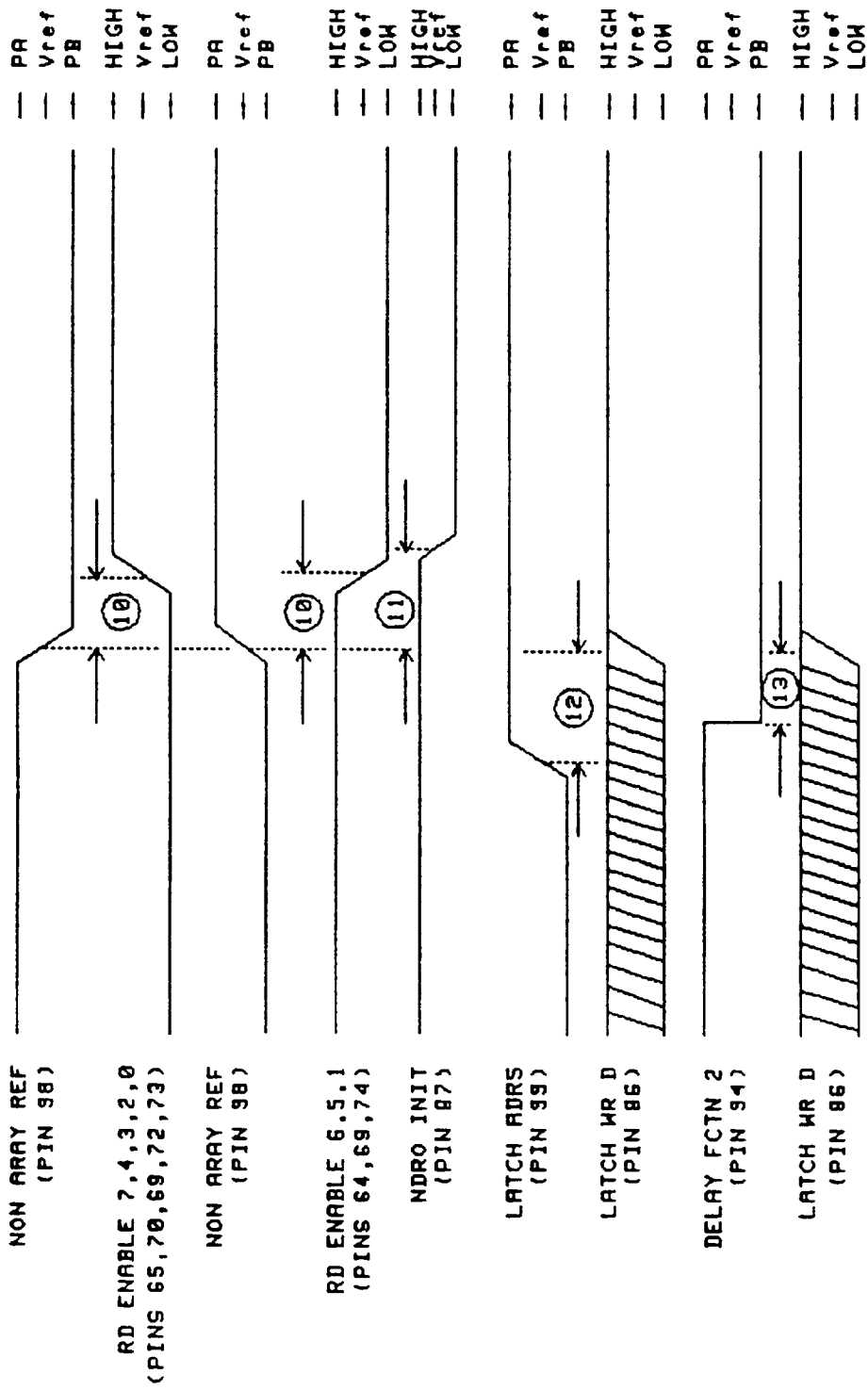


FIGURE 11c. Dynamic waveforms for table XVII.

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a. NAME Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-1528	(2) AUTOVON 88-482-1528
c. ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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